

REPUBLIC OF CAMEROON

Peace – Work – Fatherland

MINISTRY OF SECONDARY EDUCATION

INSPECTORATE GENERAL OF EDUCATION

Inspectorate of Pedagogy in charge of the
Teaching of Computer Science



REPUBLIQUE DU CAMEROUN

Paix – Travail – Patrie

MINISTÈRE DES ENSEIGNEMENTS SECONDAIRES

INSPECTION GÉNÉRALE DES ENSEIGNEMENTS

Inspection de Pédagogie chargée de
l'enseignement de l'informatique

NATIONAL HARMONISED PROGRESSION SHEET FOR COMPUTER SCIENCE FORM 3

SCHOOL YEAR.....

SCHOOL.....

WEEKLY WORKLOAD: 3 periods

TEACHER.....

GRADE.....

TEL.....

Term	Week	Module	Category of action	Competency statement	Lesson title	Objectives	Resources	Nature of lesson		
								Th	Prac	Dig
FIRST TERM	1	Hardware and software systems 1	Describing how computing technology is used in different areas of life	Given sectors of life or a situation related to application areas of computing technology, learners describe how computing technology is used in different sectors of life. The description should be coherent with the situation and sector of life and should entail the technology used, how it is used, and the advantages of using such technology.	Lesson 1: Introduction to Artificial Intelligence	- Define AI - Explain the history and evolution of AI. - Identify and categorise the different types of AI in real world				
					Lesson 2: Applications of Artificial Intelligence in Society	- Identify everyday instances where AI technology is actively used (medicine, business, agriculture, ...). - State at least two advantages of integrating AI into a given sector of life. - Describe how an AI system processes real-world data (inputs) to make decisions or predictions.				
					Lesson 3: Societal Impact and Challenges of Applied AI	- Compare the efficiency and limitations of an AI-driven system versus a traditional human-led process in a specific sector. - Explain ethical challenges or disadvantages of applied AI (e.g., data privacy risks, algorithmic bias, or job displacement).				
	2				Lesson 4: The use of appropriate Prompts to generate AI responses	- Examine prompt results - Discuss prompt results - Write appropriate prompts to solve problems within a given context				
					Lesson 5: Robots and their applications	- Explain the concepts of robot, and robotics. - Identify the key physical characteristics and				

						- roles of a robot in a given scenario. - Describe how robots are applied in specific sectors of life (manufacturing, agriculture, medicine, ...) and the advantages they bring over human labour.				
					Lesson6: Applications of control systems, virtual reality and augmented reality	- Distinguish between a monitoring system, a control system, Virtual Reality (VR), and Augmented Reality (AR). - Describe how these technologies are applied to solve real-world problems in sectors like education, aviation, and medicine.				
					Lesson 7: Integration activities					
	3		Describing internal components of the computer	Provided with a situation with factors related to devices that are not peripherals, learners select devices or features of these devices that are coherent to the situation and justify their choice.	Lesson 8: The processor	- Explain the concepts of CPU, GPU, processor core, multicore and clock speed. - Identify units for measuring the speed of the processor. - Compare mobile device processors with personal computer processors.				
					Lesson9: The processor cycle	- Describe the components of the CPU. - Explain the machine instruction cycle.				
					Lesson 10: Primary storage	- Explain the concepts of storage, primary storage, volatile, and non-volatile storage. - Identify units for measuring storage. - State the functions of cache, RAM, and register.				
					Lesson 11: Secondary storage	- Explain the concept of secondary storage. - Describe common types of secondary storage. - Compare HDD and SSDs.				
					Lesson 12: Other Internal components	- State the function of the motherboard. - State the function of power supply, battery and fans in a computer. - Describe the different types of buses.				
					Lesson 13: Computer ports	Describe the main ports of the motherboard of a computer and mobile device.				
					Lesson 14: Integration activities					
					Evaluation					
					Remediation					
	4		Describe data capture technologies	Given a situation with issues related to data capture, learners choose the most appropriate data capture technology clearly justifying their choice.	Lesson 15: Data capture technologies 1	- Explain the concept of data capture - Outline advantages and disadvantages of using data capture technologies - Describe data capture technologies				
					Lesson 16: Data capture Technologies 2	- Describe data capture technologies. - Compare and contrast data capture				
5										

						technologies.				
					Lesson 17: Integration activities					
	7	Problem solving and coding 1	Exploring basic algorithmic instructions for computers	Given an algorithmic problem, learners construct a model that solves the problem and determine simple algorithmic instructions needed to solve the problem.	Lesson 18: Program development life cycle	• Explain the concept of program development • Describe the stages in the program development life cycle. • Apply the program development life cycle to a problem				
					Lesson 19: Decomposition and representation of solutions	• Decompose problems into its component parts. • Design and construct solutions to problems using structured diagrams, and pseudocodes.				
					Lesson 20: Notions on algorithms	• Explain the concept of algorithms. • Explain characteristics of a good algorithm. • Describe simple algorithmic instructions.				
	8				Lesson 21: Algorithmic instructions	• Describe simple algorithmic instructions • Explain the purpose of a given algorithm				
					Lesson 22: Flowcharts	• Explain the concept of flowcharts • Represent solutions to problems as flowcharts				
					Lesson23: Integration activities					
	9	Hardware and software system	Choosing application software	Provided with tasks to produce or edit digital content, learners choose the most appropriate software for each task clearly justifying their choice.	Lesson24: Common application software 1	• Explain the concept of software, system software, and application software. • State the purpose of a given application software and give examples. • Outline key features of a given application. Software.				
					Lesson25: Common application software 2	• State the purpose of a given application software. • Outline key features of a given application software. • State examples of a given application software.				
					Lesson 26: Common application software 3	• State the purpose of a given application software. • Outline key features of a given application software. • State examples of a given application software.				
	10				Lesson 27: integration activities					
					Evaluation					
	11				Remediation					

SECOND TERM		12	Ethics, society and legal issues ¹	Examining licenses and copyright practices	Placed in a situation with issues related to copyright and intellectual property, learners identify and describe rules of copyright and licenses that apply to information or digital content.	Lesson 28: Protection of intellectual property	• Explain the concepts of intellectual property, copyright, unauthorise use, and fair use. • Identify situations of fair use. • Identify situations involving copyright infringement.				
						Lesson 29: Digital rights management	• Explain the meaning of digital rights management. • Outline possibilities provided by digital rights management.				
						Lesson 30: Creative Common licenses	• Explain the concept of licenses. • Describe the different types of creative common licenses.				
						Lesson 31: Integration activities					
		13	Hardware and software systems 1	Describe ways of acquiring or distributing software	Given a situation with issues involving the acquisition or distribution of software, learners recommend pertinent acquisition or distribution means while justifying their proposals clearly.	Lesson32: Off-the-shelf and Bespoke software	• Describe generic, special purpose and tailor-made software. • Give examples of generic purpose, specific and tailor-made software				
						Lesson 33: Common ways of distributing software	• Describe the following forms of software: open, closed, shareware, freeware, and software as a service. • State examples of open, closed, shareware, freeware, and software-as-a-service				
						Lesson 34: Software installation	• Differentiate between installing and updating software. • Perform software installation and update (smartphone and desktop computers).				
						Lesson 35: Integration activities					
		14		Creating digital content using software	Given tasks to accomplish, learners use appropriate features of a software to correctly accomplish the task.	Lesson 36: Edit text using a word processor	• Identify key editing features. • Perform document editing using a word processor.				
						Lesson 37: Format text using a word processor	• Identify key formatting features. • Perform document formatting using a word processor.				
						Lesson 38: Manipulate text boxes using a word processor	• Outline situations where a text box can be used. • Identify key formatting and editing actions on a text box. • Create and manipulate a text box.				
						Lesson 39: Notions on spreadsheets	• Explain the concepts of sheet, cell, address, formula, functions, and range. • Identify the address of a cell.				
						Evaluation					
						Remediation					
	15										

	16	1			Lesson 40: Basic spreadsheet functions	• Perform calculations using basic spreadsheet functions.				
					Lesson 41: Filter and chart data	• State situations where data filtering and sorting can be necessary. • Perform data filtering and sorting using a spreadsheet. • Represent cross sections of data as charts using a spreadsheet.				
					Lesson42: Notions on presentation software	• Explain the concepts of slides, animations. • Add and manipulate slides using presentation software.				
	17				Lesson 43: Animate objects in a presentation	• Identify situations where an animation can be necessary. • Create animations using presentation software.				
					Lesson 44: Using a desktop publication software	• Produce a given publication from a template. • Make use of layout features of a desktop publisher.				
					Lesson 45: Notions on Generative Artificial Intelligence	• Explain the concept of generative AI, prompt, and hallucination. • Classify generative AI tools based on their output type (text, image, video, ...). Identify core components of a prompt.				
	18				Lesson 46: Using Generative Artificial intelligence to create digital content	• Write prompt to generate a specific piece of digital content (publication, presentation, word document, ...). • Modify an initial AI-generated output by applying follow-up prompts. • Transfer, edit and format AI generated content in the appropriate traditional tool.				
					Lesson 47: Integration activities					
					19	Exploring programming languages and tools	Given a situation with factors related to programming languages, learners choose the most appropriate type of language clearly justifying their choice. Learners also demonstrate use of the main features of an IDE.	Lesson 48: Types of programming language	• Differentiate between an algorithm and a program. • Describe different types of programming languages giving an example of each. Compare high level language and low level language.	
	Lesson 49: Language translators and IDE	• Explain the concept of language translators, and IDE. • Differentiate between compiler and interpreter. Explain the role of IDEs in writing programs.								
	Lesson 50: Integration activities									
			Implementing	Given an algorithm and	Lesson 51: Introduction	• Explain the concept of programming, keywords,				

		20	algorithms using a programming language	appropriate guidance on a programming language, learners transform the algorithm into a program that is syntactically correct.	to programming	syntax, and bug. Identify the structure of a program in a programming language.				
					Evaluation					
					Lesson 52: Write, compile and execute a program 1 (python)	- Outline examples of data types, input and output instructions in a programming language. - Write programs that make use of declarative, input and output instructions.				
					Lesson 53: Write, compile and execute a program 2 (Python)	- Outline examples of operators, and assignment instructions in a programming language - Write programs that make use of declarative, input, output and assignment instructions.				
					Remediation					
					Lesson 54: Write, compile and execute a program 3 (Python)	Produce a program when given an algorithm				
	21				Lesson 55: Integration activities					
	22	Ethics, society and legal issues 1	Describing computer crimes and measures to combat them	Given a situation involving an unethical or illegal action with computers, learners identify crimes and propose appropriate measures to combat each crime.	Lesson 56: Types of computer crimes	- Explain the concept of computer crime. - Describe different types of computer crime.				
					Lesson 57: Measures to combat computer crimes	- Explain ways of fighting a given computer crime. - Explain how authentication, access control, encryption and digital forensics can help fight computer crimes.				
					Lesson 58: Authentication methods	- Explain the concept of authentication. - Describe different authentication methods.				
					Lesson 59: Integration activities					
	23	Network systems 1	Describing data communication systems and their various mechanisms	Given a situation with factors related to data communication systems, learners explain correctly, concisely, and precisely the mechanisms used in the situation.	Lesson 60: Notions of communication systems	- Define a communication system. - Identify components of a communication system. - Differentiate between digital and analogue signals				
					Lesson 61: Wired Transmission mediums	- Describe the different types of wired mediums. - Explain the concepts of bandwidth and data throughput.				
					Lesson 62: Wireless transmission mediums	Describe the different types of wireless mediums.				
					Lesson 63: Transmission mechanisms	- Describe serial and parallel transmission. - Describe simplex, half-duplex and duplex communication systems. - Describe synchronous and asynchronous transmission.				
	24				Lesson 64: Integration					

					activities					
THIRD TERM	25		Setting up simple LANs	Given a situation that requires setting up a simple network, learners prescribe the appropriate tools and configuration to meet the needs of the situation.	Evaluation					
					Remediation					
					Lesson 65: Introduction to networks	- Explain the concept of computer networks. - State advantages and disadvantages of computer networks. - Describe LAN, MAN, and WAN.				
					Lesson 66: Network devices and architecture	- Explain the function of a given network device. - Differentiate between client server and peer to peer architecture.				
					Lesson 67: Network topologies	- Explain the concept of network topology. - Describe bus, ring, and star topology. - Outline advantages and disadvantages of bus, ring, and star topology.				
					Lesson 68: Setup an adhoc network	- Describe an adhoc network. - Set up a simple adhoc network.				
	26				Lesson 69: Integration activities					
					Lesson 70: Green computing	- Explain the concept of green computing. - State devices or technologies that promote green computing.				
					Lesson 71: Measures for promoting green computing	- State measures used by manufacturers to promote green computing. - Explain ways of reducing the environmental impact of computing.				
					Lesson 72: Integration activities					
					Lesson 73: Internet services	Describe some common internet services.				
					Lesson 74: Search engines	- State examples of search engines. - Outline the main factors that influence the results of a search. - Identify common parts of a search engine result page.				
	27	Ethics, society and legal issues 1	Explaining the impact of digital technology to the environment	Given a situation with issues related to reducing the effects of computing technology on the environment, learners explain standard ways of reducing its effects that are coherent with the situation.	Lesson 75: Make use of advanced features of a search engine	- Give the purpose of a given advanced feature of a search engine. - Use advanced features of a search engine.				
					Lesson 76: Integration activities					
					Lesson 77: Introduction to web programming	- State the role of HTML and CSS in web development. - Outline the tools needed to create and view a web page. - Explain the concepts of element, tag, attribute,				
	28	Network systems 1	Searching information	Given a problem that demands the use off a search engine to find information, learners produce a logical search query and use advance features of a given search engine to ameliorate the results of a search.						
	29		Web programming							

						content, and describe the structure of an HTML page.						
	30	Problem solving and coding 1			Lesson 78: Writing HTML code 1	- Outline best practices when organising content used to build HTML pages. - Make use of the <p>, <h1> ... <h6>. - Make use of , <a> in web pages using relative paths.						
					Lesson 79: Writing HTML code 2	- Explain the necessity of nesting elements correctly and mastering the difference between inline and block elements. - State the role of the , <i>, <u>,
, <sub>, <sup> tags. - Make use of the , <i>, <u>,
, <sub>, <sup> tags in a web page.						
					Lesson 80: Writing HTML code 3	- State the role of the , , and tags - Make use of , , and tags in a web page - State the role of the different table tags - Make use of table tags in a web page						
	31				Lesson81: Introduction to CSS	- Explain the importance of CSS in web page design and development. - Make use of CSS rules to style an HTML content. - Make use of a given CSS selector to style and HTML page.						
					Lesson 82: Common CSS properties and values	- Identify when to use a given CSS property - Make use of CSS properties to format text, background and borders.						
					Lesson 83: integration activities							
	32				Evaluation							
					Remediation							
	33			Revision								
	34											
	35											
	36	END OF PROGRAM										